

Assignment 1 Report: UC Merced Land Use Classification

Course: Generative AI (GAI4UE) - SS 2026

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Dataset Preparation

- **Total images:** 2,100 (21 classes $\times$ 100 images each)
- **Split:** 70% train (1,470), 15% val (315), 15% test (315)
- **Method:** Stratified sampling

Training Results

Model 1: From Scratch (Random Initialization)

Metric	Epoch 1	Epoch 10	Epoch 20
Train Loss	3.25	2.66	2.28
Val Loss	3.14	3.22	2.15
Train Acc	6.5%	20.2%	28.5%
Val Acc	5.4%	27.0%	38.4%

Test Accuracy: 31.11%

Test Balanced Accuracy: 31.11%

Model 2: Pretrained (ImageNet Weights)

Metric	Epoch 1	Epoch 5	Epoch 20
Train Loss	3.03	0.41	0.11
Val Loss	1.57	0.20	0.05
Train Acc	23.3%	88.2%	96.7%
Val Acc	54.3%	93.0%	98.4%

Test Accuracy: 96.83%

Test Balanced Accuracy: 96.83%

Comparison

Performance Gap

- Pretrained outperforms scratch by **65.72 percentage points** (96.83% vs 31.11%)
- Pretrained reaches 93% val accuracy at epoch 5
- Scratch model never exceeds 40% val accuracy

Loss Convergence

- **Scratch:** Final loss 2.28 (train) / 2.15 (val) - still high
- **Pretrained:** Final loss 0.11 (train) / 0.05 (val) - converged well

Class-Specific Performance (Pretrained Model)

Based on confusion matrix and classification report: - **Perfect classes:** agricultural, airplane, beach, golfcourse, tenniscourt (100% F1-score) - **Challenging pairs:** Similar visual features cause confusion between residential density levels and vegetation types

Conclusions

Transfer learning with ImageNet pretrained weights is essential for this task. The scratch model failed to learn meaningful features from 1,470 training images, while the pretrained model achieved near-perfect accuracy by epoch 5. The pretrained weights provided learned edge, texture, and shape detectors that transferred well to aerial imagery classification.